

Initial Management of Noncastrate Advanced, Recurrent, or Metastatic Prostate Cancer: ASCO Guideline Update

Katherine S. Virgo, PhD, MBA¹ ; R. Bryan Rumble, MSc² ; and James Talcott, MD, SM³ 

DOI <https://doi.org/10.1200/JCO.23.00155>

BACKGROUND

In 2021, ASCO updated a guideline on the initial management of noncastrate recurrent, advanced, or metastatic prostate cancer.¹ Several recent developments prompted this focused update. First, at the time of the 2021 update, the results from the ARASENS² trial were not yet available and darolutamide was not yet US Food and Drug Administration–approved for patients with noncastrate de novo metastatic prostate cancer. However, on August 5, 2022, darolutamide, in combination with docetaxel and androgen-deprivation therapy (ADT; triplet therapy), was US Food and Drug Administration–approved for these patients based on the results of the ARASENS trial² and is one of several reasons an update was deemed necessary.

The second major reason for a focused recommendation update was publication of the results from the PEACE-1 trial³ of docetaxel plus abiraterone plus ADT (triplet therapy) among patients with de novo metastatic noncastrate prostate cancer.

The third reason for a focused recommendation update was updated results from two enzalutamide trials (ENZAMET⁴ and ARCHES⁵). The ENZAMET trial released (in abstract form only) updated survival results for patients with metastatic noncastrate prostate cancer after three additional years of follow-up.⁴ These longer-term results examine whether the previously reported early short-term survival benefits of enzalutamide plus ADT (compared with bicalutamide, nilutamide, or flutamide plus ADT) are sustained and whether there is a survival benefit among those receiving concurrent docetaxel (triplet therapy).

The related ARCHES⁵ trial published updated radiographic progression-free survival results after a longer period of follow-up and also reported results for the trial's secondary outcome, overall survival (OS) for patients with metastatic noncastrate prostate cancer, primarily those with de novo metastatic disease. This study compared enzalutamide plus ADT versus placebo plus ADT, stratified by disease volume and prior docetaxel use (sequential treatment).

Only clinical trials central to the discussion of triplet therapy and trials, the results of which necessitated changes to the recommendations, were reviewed in detail. Therefore, trials such as Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy (STAMPEDE) Arm G (abiraterone), SWOG 1216 (orteronel), and the updated TITAN (apalutamide) results are not discussed.

Thus, the current update is primarily driven by new clinical trial results comparing triplet therapies with standard of care (SoC) as well as one meta-analysis of triplet therapy.⁶ Further informing this update is one network meta-analysis comparing ADT plus an androgen receptor axis–targeted agent to ADT plus docetaxel,⁷ a comparison not yet directly tested in clinical trials but of great interest.

See Data Supplement 6 (online only): Included Trials for a summary of included trials, Appendix Table A1 (online only) for all recommendations, including those that were not updated, Data Supplement 7: Table Comparing Original Recommendations with Updated Recommendations, and Figure 1 for a decision tree depicting initial management of noncastrate metastatic prostate cancer.

ACCOMPANYING CONTENT

 Appendix

 Data Supplement

Accepted February 16, 2023

Published April 3, 2023

J Clin Oncol 41:3652-3656

© 2023 by American Society of
Clinical Oncology



View Online
Article

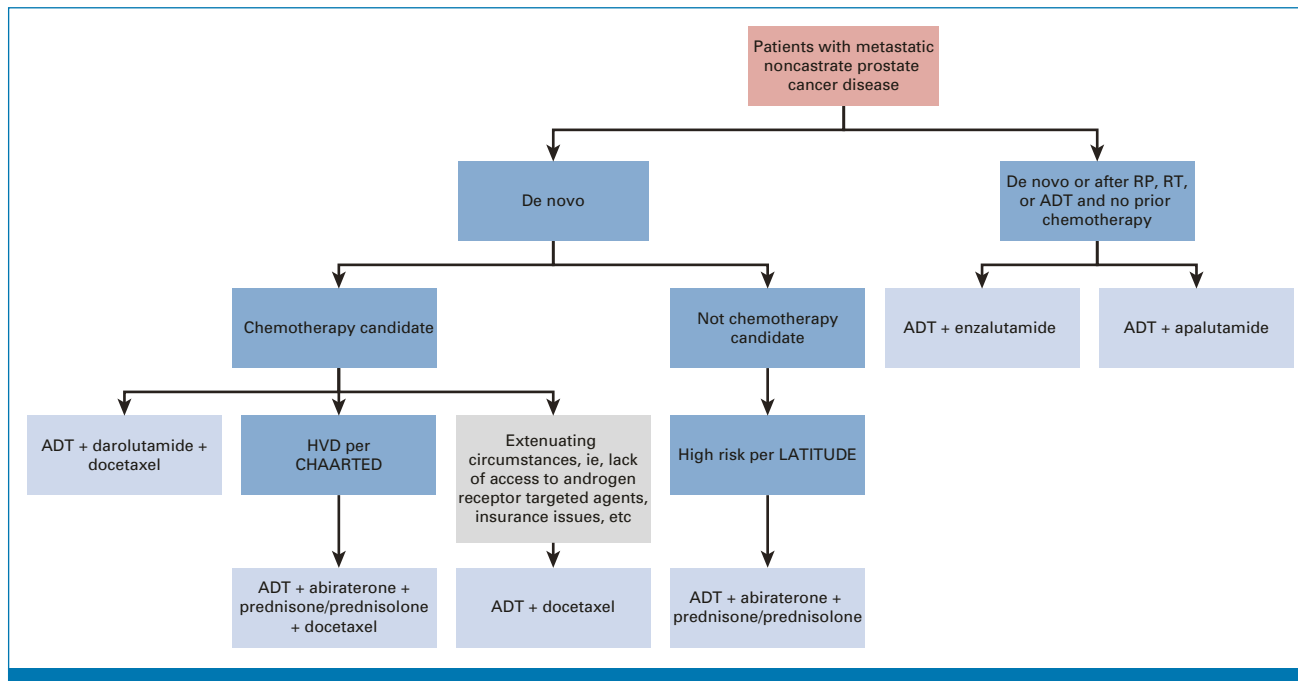


FIG 1. Initial management of noncastrate metastatic prostate cancer algorithm. ADT, androgen-deprivation therapy; HVD, high-volume disease; RP, radical prostatectomy; RT, radiotherapy.

METHODS

This systematic review-based guideline product was developed by a multidisciplinary Expert Panel, which included a patient representative and an ASCO guidelines staff member with health research methodology expertise. Evidence supporting unchanged recommendations is reviewed in the previous versions of this guideline.¹ See Data Supplement 3: Methods (expanded) for more information. All funding for the administration of the project was provided by ASCO.

EVIDENCE REVIEW

See Data Supplement 4: Evidence Review (expanded) for full descriptions of the included trials, including dosing, inclusion criteria, and outcomes.

UPDATED RECOMMENDATIONS

Recommendation 1.0

Docetaxel, abiraterone, enzalutamide, apalutamide, or darolutamide each when administered with ADT, represent five separate SoCs for noncastrate metastatic prostate cancer. With the exception of the triplet therapies of docetaxel plus abiraterone plus ADT and docetaxel plus darolutamide plus ADT, the use of any of these agents in any other particular combination or in any particular series cannot yet be recommended (Type: Evidence-based, benefits-harms ratio unknown; Evidence quality: No evidence available; Strength of recommendation: Strong).

Recommendation 1.1

For patients with metastatic noncastrate prostate cancer with high-volume disease (HVD) as defined per CHAARTED⁸ (four or more bone metastases, one or more of which is outside of the spine or pelvis, and/or the presence of any visceral disease) who are candidates for treatment with chemotherapy but are unwilling or unable to receive triplet therapy (eg, due to insurance constraints), docetaxel plus ADT should be offered (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with HVD).

Practical Information for Recommendation 1.1

Patients should be made aware that doublet therapy (docetaxel plus ADT) has proven inferior OS compared to triplet therapy, such as abiraterone and prednisone plus docetaxel plus ADT.

Recommendation 1.11

For patients with de novo metastatic noncastrate prostate cancer with HVD as defined per CHAARTED⁸ who are being offered ADT plus docetaxel chemotherapy, triplet therapy (abiraterone and prednisone plus ADT and docetaxel) should be offered per PEACE-1.³ Abiraterone and prednisone plus ADT and docetaxel demonstrated significant OS and radiographic progression-free survival benefits compared to ADT and docetaxel alone for patients with HVD (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with HVD as defined per CHAARTED).⁸

Practical Information for Recommendation 1.11

OS data for patients with low-volume de novo metastatic noncastrate prostate cancer from PEACE-1³ are still too immature to justify recommending abiraterone-based triplet therapy (abiraterone and prednisone plus ADT and docetaxel).

Recommendation 1.15

For patients with de novo metastatic noncastrate prostate cancer who are being offered ADT plus docetaxel chemotherapy, triplet therapy (darolutamide plus ADT and docetaxel) should be offered per ARASENS.² Compared to placebo plus ADT and docetaxel, darolutamide plus ADT and docetaxel demonstrated significant OS benefits, in addition to significantly longer times to castration-resistant prostate cancer, pain progression, first skeletal event, and initiation of subsequent systemic antineoplastic therapy (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 1.16

The recommended regimen for patients with metastatic noncastrate prostate cancer treated with darolutamide, docetaxel and ADT is darolutamide (600 mg as two 300 mg tablets orally with food) twice daily (to a total daily dose of 1200 mg) with ADT. Docetaxel administration (75 mg/m²) should begin within 6 weeks of the first dose of darolutamide. Docetaxel should be administered intravenously once every 3 weeks for up to six cycles (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Practical Information for Recommendations 1.15 and 1.16

Discussions with patients should include the cost of darolutamide treatment compared with other options such as abiraterone.

Recommendation 1.2

For patients with low-volume metastatic disease (LVD) as defined per CHAARTED⁸ who are candidates for chemotherapy, docetaxel plus ADT should not be offered due to lack of sufficient evidence (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with LVD).

Recommendation 1.3

For patients with metastatic noncastrate prostate cancer treated with docetaxel, the recommended regimen is six doses administered once at 3-week intervals at 75 mg/m² either alone (per CHAARTED⁸) or with prednisolone (per STAMPEDE⁹; Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Practical Information for Recommendation 1.3

- The strongest evidence of benefit for docetaxel is for those patients who were diagnosed with de novo metastatic disease or HVD (defined per CHAARTED⁸ as four or more bone metastases, one or more of which is outside of the spine or pelvis, and/or the presence of any visceral disease). The criteria apply independent of the presence or absence of nodal disease.¹⁰
- Patients with metastatic disease who do not fit into these categories should not be offered docetaxel. The strength of the evidence to support an OS benefit is not compelling for patients who do not have de novo metastatic disease and/or who do not meet the HVD criteria.¹⁰ Long-term survival data from CHAARTED⁸ and a post hoc aggregated analysis of CHAARTED and GETUG-AFU-15 data only showed an OS benefit for patients with HVD and de novo metastases. There was no OS benefit for LVD, irrespective of whether the patients had metastases at diagnosis or after failure of prior local therapy.¹¹ Clarke et al¹² re-examined OS by disease burden using STAMPEDE data with longer follow-up, but the analysis had inadequate statistical power (< 80%) to detect an OS difference by disease burden if in fact one existed.
- As a chemotherapy agent, docetaxel is associated with somewhat greater toxicity than androgen-targeted therapies (novel hormone therapies), but the treatment course is relatively short (limited to six cycles), much less costly, and generally covered by insurance, hence reducing the financial toxicity.

ADT Plus Enzalutamide**Recommendation 1.7**

ADT plus enzalutamide should be offered to patients with metastatic noncastrate prostate cancer including both those with de novo metastatic disease and those who have received prior therapies, such as radical prostatectomy or radiotherapy for localized disease. Enzalutamide plus ADT has demonstrated short-term survival benefits (prostate specific antigen progression-free, clinical progression-free, and overall) when compared with ADT alone for patients with metastatic noncastrate prostate cancer as a group per ENZAMET.¹³ Enzalutamide also has long-term survival benefits overall, for those with LVD or HVD, and those with LVD or HVD and no docetaxel use.⁴ Per ARCHES,⁵ enzalutamide significantly improved time to first subsequent antineoplastic therapy in addition to survival benefits overall and among those with HVD, no prior docetaxel and previous use of ADT or orchiectomy⁵ (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 1.8

The recommended regimen for patients with metastatic noncastrate prostate cancer is enzalutamide (160 mg once

per day) with ADT (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Practical Information for Recommendation 1.8

Discussions with patients should include the cost of enzalutamide treatment compared with other options, such as abiraterone.

Practical Information for Recommendations 1.9 and 1.95

Discussions with patients should include the cost of apalutamide treatment compared with other options, such as abiraterone.

Additional information including a supplement, and clinical tools and resources can be found at www.asco.org/genitourinary-cancer-guidelines. Patient information is available there and at www.cancer.net.

Guideline Disclaimer

The Clinical Practice Guidelines and other guidance published herein are provided by the ASCO to assist providers in clinical decision making. The information herein should not be relied upon as being complete or accurate, nor should it be considered as inclusive of all proper treatments or methods of care or as a statement of the SoC. With the rapid development of scientific knowledge, new evidence may emerge between the time information is developed and when it is published or read. The information is not continually updated and may not reflect the most recent evidence. The information addresses only the topics specifically identified therein and is not applicable to other interventions, diseases, or stages of diseases. This information does not mandate any particular course of medical care. Further, the information is not intended to substitute for the independent professional judgment of the treating provider, as the information does not account for individual

variation among patients. Recommendations specify the level of confidence that the recommendation reflects the net effect of a given course of action. The use of words like “must,” “must not,” “should,” and “should not” indicates that a course of action is recommended or not recommended for either most or many patients, but there is latitude for the treating physician to select other courses of action in individual cases. In all cases, the selected course of action should be considered by the treating provider in the context of treating the individual patient. Use of the information is voluntary. ASCO does not endorse third party drugs, devices, services, or therapies used to diagnose, treat, monitor, manage, or alleviate health conditions. Any use of a brand or trade name is for identification purposes only. ASCO provides this information on an “as is” basis and makes no warranty, express or implied, regarding the information. ASCO specifically disclaims any warranties of merchantability or fitness for a particular use or purpose. ASCO assumes no responsibility for any injury or damage to persons or property arising out of or related to any use of this information, or for any errors or omissions.

Guideline and Conflicts of Interest

The Expert Panel was assembled in accordance with ASCO’s Conflict of Interest Policy Implementation for Clinical Practice Guidelines (“Policy,” found at <https://www.asco.org/guideline-methodology>). All members of the Expert Panel completed ASCO’s disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker’s bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

AFFILIATIONS

¹Department of Health Policy and Management, Rollins School of Public Health, Emory University, Atlanta, GA

²American Society of Clinical Oncology, Alexandria, VA

³NantHealth Inc, Culver City, CA

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Rd, Ste 800, Alexandria, VA 22314; e-mail: guidelines@asco.org.

EDITOR’S NOTE

This ASCO Clinical Practice Guideline Update provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence, slide sets, clinical

tools and resources, and links to patient information at www.cancer.net, is available at www.asco.org/genitourinary-cancer-guidelines.

EQUAL CONTRIBUTION

K.S.V. and J.T. were expert panel co-chairs.

AUTHOR’S DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the author are available with this article at DOI <https://doi.org/10.1200/JCO.23.00155>.

AUTHOR CONTRIBUTIONS

Conception and design: All authors
Administrative support: R. Bryan Rumble
Collection and assembly of data: All authors
Data analysis and interpretation: All authors
Manuscript writing: All authors
Final approval of manuscript: All authors
Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

The authors wish to thank Dr A. Schuckman, MD, FACS, and Dr M.P. Mullane, MD, and the ASCO Evidence-Based Medicine Committee for

their thoughtful reviews and insightful comments on this guideline update. The following are members of the Initial Management of Noncastrate Advanced, Recurrent, or Metastatic Prostate Cancer Expert Panel: Katherine S. Virgo, PhD, MBA; R. Bryan Rumble, MSc; Ronald de Wit, MD, PhD; Thomas J. Smith, MD; Mary-Ellen Taplin, MD; James L. Wade III, MD; Charles L. Bennett, MD, PhD; Howard I. Scher, MD; Paul L. Nguyen, MD; Martin Gleave, MD; Scott C. Morgan, MD, MSc; Andrew Loblaw, MD, MSc; Sean Sachdev, MD; David L. Graham, MD; Neha Vapiwala, MD; Winston Tan, MD; Terry M. Kungel, MBA; and James A. Talcott, MD, SM. The Update Panel and ASCO staff dedicate this guideline to the memory of Nicholas J. Vogelzang, MD, our dearly departed colleague and former panel member.

REFERENCES

1. Virgo KS, Rumble RB, de Wit R, et al: Initial management of noncastrate advanced, recurrent, or metastatic prostate cancer: ASCO guideline update. *J Clin Oncol* 39:1274-1305, 2021
2. Smith MR, Hussain M, Saad F, et al: Darolutamide and survival in metastatic, hormone-sensitive prostate cancer. *N Engl J Med* 386:1132-1142, 2022
3. Fizazi K, Foulon S, Carles J, et al: Abiraterone plus prednisone added to androgen deprivation therapy and docetaxel in de novo metastatic castration-sensitive prostate cancer (PEACE-1): A multicentre, open-label, randomised, phase 3 study with a 2 × 2 factorial design. *Lancet* 399:1695-1707, 2022
4. Davis ID, Martin AJ, Zielinski RR, et al: Updated overall survival outcomes in ENZAMET (ANZUP 1304), an international, cooperative group trial of enzalutamide in metastatic hormone-sensitive prostate cancer (mHSPC). *J Clin Oncol* 40, 2022 (17 suppl; abstr LBA5004)
5. Armstrong AJ, Iguchi T, Azad A, et al: Overall survival (OS) in patients (pts) with metastatic hormone-sensitive prostate cancer (mHSPC) treated with enzalutamide (ENZA)+ androgen deprivation therapy (ADT) by high or low disease volume and progression to mHSPC (M0 at diagnosis) or de novo mHSPC (M1 at diagnosis): Post hoc analysis of the phase 3 ARCHES trial. *J Clin Oncol* 40, 2022 (6 suppl; abstr 115)
6. Ciccarese C, Iacovelli R, Sternberg CN, et al: Triplet therapy with androgen deprivation, docetaxel, and androgen receptor signalling inhibitors in metastatic castration-sensitive prostate cancer: A meta-analysis. *Eur J Cancer* 173:276-284, 2022
7. Ferro M, Lucarelli G, Crocetto F, et al: First-line systemic therapy for metastatic castration-sensitive prostate cancer: An updated systematic review with novel findings. *Crit Rev Oncol Hematol* 157: 103198, 2021
8. Sweeney CJ, Chen YH, Carducci M, et al: Chemohormonal therapy in metastatic hormone-sensitive prostate cancer. *N Engl J Med* 373:737-746, 2015
9. James ND, Sydes MR, Clarke NW, et al: Addition of docetaxel, zoledronic acid, or both to first-line long-term hormone therapy in prostate cancer (STAMPEDE): Survival results from an adaptive, multiarm, multistage, platform randomised controlled trial. *Lancet* 387:1163-1177, 2016
10. Morris MJ, Rumble RB, Basch E, et al: Optimizing anticancer therapy in metastatic non-castrate prostate cancer: American Society of Clinical Oncology clinical practice guideline. *J Clin Oncol* 36: 1521-1539, 2018
11. Kyriakopoulos CE, Chen Y-H, Carducci MA, et al: Chemohormonal therapy in metastatic hormone-sensitive prostate cancer: Long-term survival analysis of the randomized phase III E3805 CHAARTED trial. *J Clin Oncol* 36:1080-1087, 2018
12. Clarke NW, Ali A, Ingleby F, et al: Addition of docetaxel to hormonal therapy in low- and high-burden metastatic hormone sensitive prostate cancer: Long-term survival results from the STAMPEDE trial. *Ann Oncol* 30:1992-2003, 2019
13. Davis ID, Martin AJ, Stockler MR, et al: Enzalutamide with standard first-line therapy in metastatic prostate cancer. *N Engl J Med* 381:121-131, 2019
14. Fizazi K, Tran N, Fein L, et al: Abiraterone plus prednisone in metastatic, castration-sensitive prostate cancer. *N Engl J Med* 377:352-360, 2017
15. James ND, de Bono JS, Spears MR, et al: Abiraterone for prostate cancer not previously treated with hormone therapy. *N Engl J Med* 377:338-351, 2017
16. Chi KN, Agarwal N, Bjartell A, et al: Apalutamide for metastatic, castration-sensitive prostate cancer. *N Engl J Med* 381:13-24, 2019
17. Magnan S, Zarychanski R, Pilote L, et al: Intermittent vs continuous androgen deprivation therapy for prostate cancer: A systematic review and meta-analysis. *JAMA Oncol* 1:1261-1269, 2015
18. Botrel TEA, Clark O, Dos Reis RB, et al: Intermittent versus continuous androgen deprivation for locally advanced, recurrent or metastatic prostate cancer: A systematic review and meta-analysis. *BMC Urol* 14:9, 2014
19. Brungs D, Chen J, Masson P, et al: Intermittent androgen deprivation is a rational standard-of-care treatment for all stages of progressive prostate cancer: Results from a systematic review and meta-analysis. *Prostate Cancer Prostatic Dis* 17:105-111, 2014
20. Niraula S, Le LW, Tannock IF: Treatment of prostate cancer with intermittent versus continuous androgen deprivation: A systematic review of randomized trials. *J Clin Oncol* 31:2029-2036, 2013
21. Tsai H-T, Penson DF, Makambi KH, et al: Efficacy of intermittent androgen deprivation therapy vs conventional continuous androgen deprivation therapy for advanced prostate cancer: A meta-analysis. *Urology* 82:327-334, 2013
22. Van den Broeck T, van den Bergh RCN, Arfi N, et al: Prognostic value of biochemical recurrence following treatment with curative intent for prostate cancer: A systematic review. *Eur Urol* 75: 967-987, 2019

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Initial Management of Noncastrate Advanced, Recurrent, or Metastatic Prostate Cancer: ASCO Guideline Update

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/jco/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](#)).

James Talcott

Employment: NantHealth

Open Payments Link: <https://openpaymentsdata.cms.gov/physician/1242101/>

No other potential conflicts of interest were reported.

APPENDIX

TABLE A1. Summary of All Recommendations

Recommendation	Type, Evidence Quality, Strength of Recommendation
<i>Recommendation 1.0 (updated):</i> Docetaxel, abiraterone, enzalutamide, apalutamide, or darolutamide each when administered with ADT, represent five separate SoCs for noncastrate metastatic prostate cancer. With the exception of the triplet therapies of docetaxel plus abiraterone plus ADT and docetaxel plus darolutamide plus ADT, the use of any of these agents in any other particular combination or in any particular series cannot yet be recommended.	Type: Evidence-based, benefits-harms ratio unknown; Evidence quality: No evidence available; Strength of recommendation: Strong
ADT plus docetaxel ¹⁰ <i>Recommendation 1.1 (updated):</i> For patients with metastatic noncastrate prostate cancer with HVD as defined per CHAARTED ⁸ (four or more bone metastases, one or more of which is outside of the spine or pelvis, and/or the presence of any visceral disease) who are candidates for treatment with chemotherapy but are unwilling or unable to receive triplet therapy (eg, due to insurance constraints), docetaxel plus ADT should be offered.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with HVD
<i>Practical information for Recommendation 1.1 (updated):</i> Patients should be made aware that doublet therapy (docetaxel plus ADT) has proven inferior OS compared to triplet therapy, such as abiraterone and prednisone plus docetaxel plus ADT.	
<i>Recommendation 1.11 (updated):</i> For patients with de novo metastatic noncastrate prostate cancer with HVD as defined per CHAARTED ⁸ who are being offered ADT plus docetaxel chemotherapy, triplet therapy (abiraterone and prednisone plus ADT and docetaxel) should be offered per PEACE-1. ³ Abiraterone and prednisone plus ADT and docetaxel demonstrated significant OS and rPFS benefits compared to ADT and docetaxel alone for patients with HVD.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with HVD as defined per CHAARTED ⁸
<i>Practical information for Recommendation 1.11 (updated):</i> OS data for patients with low-volume de novo metastatic noncastrate prostate cancer from PEACE-1 ³ are still too immature to justify recommending abiraterone-based triplet therapy (abiraterone and prednisone plus ADT and docetaxel).	
<i>Recommendation 1.15 (updated):</i> For patients with de novo metastatic noncastrate prostate cancer who are being offered ADT plus docetaxel chemotherapy, triplet therapy (darolutamide plus ADT and docetaxel) should be offered per ARASENS. ² Compared to placebo plus ADT and docetaxel, darolutamide plus ADT and docetaxel demonstrated significant OS benefits, in addition to significantly longer times to castration-resistant prostate cancer, pain progression, first skeletal event, and initiation of subsequent systemic antineoplastic therapy.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Recommendation 1.16 (updated):</i> The recommended regimen for patients with metastatic noncastrate prostate cancer treated with darolutamide, docetaxel, and ADT is darolutamide (600 mg as two 300 mg tablets orally with food) twice daily (to a total daily dose of 1200 mg) with ADT. Docetaxel administration (75 mg/m ²) should begin within 6 weeks of the first dose of darolutamide. Docetaxel should be administered intravenously once every 3 weeks for up to six cycles.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Practical information for Recommendations 1.15 and 1.16 (updated):</i> Discussions with patients should include the cost of darolutamide treatment compared with other options such as abiraterone.	
<i>Recommendation 1.2 (updated):</i> For patients with LVD as defined per CHAARTED ⁸ who are candidates for chemotherapy, docetaxel plus ADT should not be offered due to lack of sufficient evidence.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with LVD
<i>Recommendation 1.3 (updated):</i> For patients with metastatic noncastrate prostate cancer treated with docetaxel, the recommended regimen is six doses administered once at 3-week intervals at 75 mg/m ² either alone (per CHAARTED ⁸) or with prednisolone (per STAMPEDE). ⁹	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Practical information for Recommendation 1.3 (updated)</i> The strongest evidence of benefit for docetaxel is for those patients who were diagnosed with de novo metastatic disease or HVD (defined per CHAARTED ⁸ as four or more bone metastases, one or more of which is outside of the spine or pelvis, and/or the presence of any visceral disease). The criteria apply independent of the presence or absence of nodal disease. ¹⁰ Patients with metastatic disease who do not fit into these categories should not be offered docetaxel. The strength of the evidence to support an OS benefit is not compelling for patients who do not have de novo metastatic disease and/or who do not meet the HVD criteria. ¹⁰ Long-term survival data from CHAARTED ⁸ and a post hoc aggregated analysis of CHAARTED and GETUG-AFU-15 data only showed an OS benefit for patients with HVD and de novo metastases. There was no OS benefit for LVD, irrespective of whether the patients had metastases at diagnosis or after failure of prior local therapy. ¹¹ Clarke et al ¹² re-examined OS by disease burden using STAMPEDE data with longer follow-up, but the analysis had inadequate statistical power (< 80%) to detect an OS difference by disease burden if in fact one existed. As a chemotherapy agent, docetaxel is associated with somewhat greater toxicity than androgen-targeted therapies (novel hormone therapies), but the treatment course is relatively short (limited to six cycles), much less costly, and generally covered by insurance, hence reducing the financial toxicity.	
ADT plus abiraterone ¹⁰ <i>Recommendation 1.4:</i> For patients with high-risk de novo metastatic noncastrate prostate cancer, the addition of abiraterone to ADT should be offered per LATITUDE. ¹⁴	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong for patients with high-risk disease as defined per LATITUDE
<i>Recommendation 1.5:</i> For patients with low-risk de novo metastatic noncastrate prostate cancer, ADT plus abiraterone may be offered per STAMPEDE. ¹⁵	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Moderate for patients with low-risk disease per STAMPEDE

(continued on following page)

TABLE A1. Summary of All Recommendations (continued)

Recommendation	Type, Evidence Quality, Strength of Recommendation
<i>Recommendation 1.6:</i> The recommended regimen for patients with metastatic noncastrate prostate cancer is abiraterone 1,000 mg once per day with either prednisolone or prednisone 5 mg once daily until progressive disease is documented.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
ADT plus enzalutamide <i>Recommendation 1.7 (updated):</i> ADT plus enzalutamide should be offered to patients with metastatic noncastrate prostate cancer including both those with de novo metastatic disease and those who have received prior therapies, such as RP or RT for localized disease. Enzalutamide plus ADT has demonstrated short-term survival benefits (PSA progression-free, clinical progression-free, and overall) when compared with ADT alone for patients with metastatic noncastrate prostate cancer as a group per ENZAMET. ¹³ Enzalutamide also has long-term survival benefits overall, for those with LVD or HVD, and those with LVD or HVD and no docetaxel use. ⁴ Per ARCHES, ⁵ enzalutamide significantly improved time to first subsequent antineoplastic therapy in addition to survival benefits overall and among those with high-volume disease, no prior docetaxel and previous use of ADT or orchiectomy. ⁵	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Recommendation 1.8 (updated):</i> The recommended regimen for patients with metastatic noncastrate prostate cancer is enzalutamide (160 mg once per day) with ADT.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Practical information for Recommendation 1.8 (updated):</i> Discussions with patients should include the cost of enzalutamide treatment compared with other options, such as abiraterone.	
ADT plus apalutamide <i>Recommendation 1.9:</i> ADT plus apalutamide should also be offered to patients with metastatic noncastrate prostate cancer, including those with de novo metastatic disease or those who have received prior therapy, such as RP or radiation therapy for localized disease per TITAN. ¹⁶	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Recommendation 1.95:</i> The recommended regimen for patients with metastatic noncastrate prostate cancer is apalutamide (240 mg once per day) with ADT.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Practical information for Recommendations 1.9 and 1.95 (updated):</i> Discussions with patients should include the cost of apalutamide treatment compared with other options, such as abiraterone.	
<i>Recommendation 2.1:</i> ADT plus abiraterone and prednisolone should be considered for patients with noncastrate locally advanced nonmetastatic prostate cancer, rather than castration monotherapy, due to the failure-free survival benefit per STAMPEDE. ¹⁵ RT to the primary was mandated in STAMPEDE for patients with newly diagnosed node-negative, nonmetastatic disease and encouraged in patients with newly diagnosed node-positive, nonmetastatic disease. Failure-free survival (time to the earliest of biochemical failure, disease progression, or death), was significantly improved for patients with nonmetastatic disease treated with ADT plus abiraterone and prednisolone compared to those treated with ADT alone, even though ADT plus abiraterone was administered for two or less years to patients with nonmetastatic disease.	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong
<i>Recommendation 2.2:</i> In resource-constrained settings where drugs such as abiraterone may not be available, combined androgen blockade using ADT plus a first-generation antiandrogen, such as flutamide, nilutamide, or bicalutamide, may be offered to patients with locally advanced nonmetastatic prostate cancer, rather than castration monotherapy based on recent meta-analyses.	Type: Evidence-based; Evidence quality: High, benefits outweigh harms; Strength of recommendation: Moderate
<i>Practical information for Recommendation 2.2:</i> For patients with high-risk nonmetastatic prostate cancer progressing after RP or RT or both, it is currently unclear whether enzalutamide (160 mg once per day) plus leuprolide improves metastasis-free survival compared to enzalutamide monotherapy or placebo. Though recruitment is complete for the ongoing phase III EMBARK trial, which is designed to answer this question, results are not yet available. Thus, no recommendation can be made at this time.	
<i>Recommendation 3.1:</i> Early (immediate) ADT may be offered to patients who initially present with noncastrate locally advanced nonmetastatic disease who have not undergone previous local treatment and are unwilling or unable to undergo RT based on evidence in one meta-analysis of a modest, but statistically significant benefit in terms of both OS and cancer-specific survival among the larger population of patients with locally advanced nonmetastatic disease.	Type: Evidence-based, benefits outweigh harms; Evidence quality: Intermediate; Strength of recommendation: Moderate
<i>Practical information for Recommendation 3.1</i> Discussions with patients regarding early ADT should include the risk of short- and long-term side effects. Deferred ADT is often preferred by patients who desire to avoid, or at least delay, potential ADT side effects. Consideration should be given to restricting deferred ADT to those patients who are asymptomatic. No recommendation can be provided at this time for patients with PSA relapse after local treatment. Though existing studies suggest a potential OS benefit, additional research is needed as such studies were underpowered.	
<i>Recommendation 4.1:</i> Intermittent therapy may be offered to patients with high-risk biochemically recurrent nonmetastatic prostate cancer after RP and/or RT based on evidence in meta-analyses of the noninferiority of IADT when compared to CADT with respect to OS. ¹⁷ This is further supported by evidence from four meta-analyses ¹⁸⁻²¹ testing superiority. Low-risk biochemical recurrence after RP is defined as a PSA doubling time > 1 year and pathologic Gleason score < 8. Low-risk biochemical	Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong

(continued on following page)

TABLE A1. Summary of All Recommendations (continued)

Recommendation	Type, Evidence Quality, Strength of Recommendation
recurrence after radiation therapy is defined as an interval to biochemical recurrence > 18 months and clinical Gleason score < 8. High-risk biochemical recurrence after RP is defined as a PSA doubling time < 1 year or a pathologic Gleason score of 8-10. High-risk biochemical recurrence after radiation therapy is defined as an interval to biochemical recurrence < 18 months or a clinical Gleason score of 8-10.²² Active surveillance may be offered to patients with low-risk biochemically recurrent nonmetastatic prostate cancer.	
<i>Practical information for Recommendation 4.1</i>	
Although patients with noncastrate de novo metastatic prostate cancer were included in the studies reviewed for this clinical question, alternative SoC therapies with proven survival benefits now exist, as outlined in Recommendation 1 to include ADT plus docetaxel, ADT plus abiraterone, ADT plus enzalutamide, or ADT plus apalutamide. Similar support for these existing standards of care does not universally exist for patients with LVD and further research is needed. No specific additional recommendation with respect to the use of IADT in the noncastrate metastatic prostate cancer population was possible at this time because IADT has not been studied in combination with additional cytotoxic or hormonal agents in this population. Patients considering IADT should be made aware of the potential benefits of IADT associated with the off-treatment intervals, such as reduced treatment side effects, quality-of-life benefits, and lower cost. As patients on IADT require close follow-up, they must be motivated to adhere to frequent doctor visits for monitoring, even during off-treatment periods.	

Abbreviations: ADT, androgen-deprivation therapy; CADT, continuous androgen-deprivation therapy; HVD, high-volume disease; IADT, intermittent androgen-deprivation therapy; LVD, low-volume metastatic disease; OS, overall survival; PSA, prostate-specific antigen; RP, radical prostatectomy; rPFS, radiographic progression-free survival; RT, radiotherapy; SoC, standard of care; STAMPEDE, Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy.